Device and Process Engineering Submission

Aluminium-Gate MOS Capacitor - FabuBlox Workflow Evidence

Scope and integrity statement

This submission documents a saved process flow and its FabuBlox simulation view. It is a design and workflow exercise, not evidence of physical fabrication or electrical testing. All material parameters and recipe settings should be reviewed against approved HackerFab IITB processes before any laboratory use.

Item	Evidence / status
Device	Aluminium-gate MOS capacitor on lightly p-doped Si with SiO2 dielectric.
FabuBlox flow	Saved project: Aluminium-Gate MOS Capacitor; 12 displayed workflow steps.
Simulation	Final captured simulation view displays 'No notices.' This is workflow simulation evidence only.
Not completed	No physical fabrication, annealing step, electrical characterization, or C-V / leakage measurements claimed.

1. Device concept

The proposed structure is a metal-oxide-semiconductor capacitor: aluminium gate / silicon dioxide dielectric / lightly p-doped silicon. In the captured FabuBlox layer stack, the dielectric is shown as 20 nm SiO2 and the aluminium layer as 100 nm. A MOS capacitor is a compact learning test structure for linking dielectric formation, metal patterning, contact design, and eventual C-V or leakage measurements.

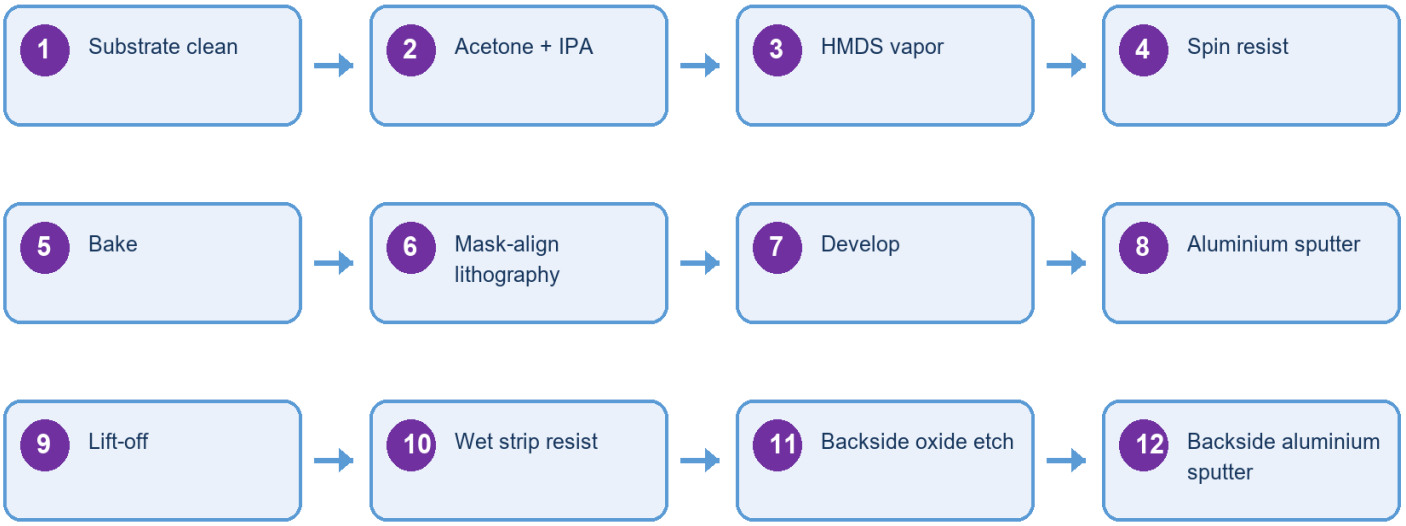
2. Saved FabuBlox workflow

The project is named **Aluminium-Gate MOS Capacitor**. The saved workflow shows substrate preparation, resist processing, lithography, aluminium sputtering, lift-off, resist stripping, backside oxide etch, and backside aluminium sputtering. The supplied screenshots are retained unchanged in **evidence/**. The supporting process summary is included below.

This is the link: https://www.fabublox.com/designer/050ab4a2-a4c1-431f-9c19-8e2e0e90a646

Aluminium-Gate MOS Capacitor: FabuBlox Process Summary

Workflow reconstructed from the saved FabuBlox flow; this is not a physical fabrication record.



Evidence and limits

The saved flow includes a Si / SiO2 / Al stack (20 nm SiO2, 100 nm Al) and a final simulation view with no notices.

Annealing and electrical characterization were unavailable in the editor and are not represented as completed steps.

3. Workflow evidence

Figures 2-5 reproduce the user's original screenshots. They show the layer stack, the first and last workflow sections, and the final simulation panel. The last panel contains the text 'No notices'; that observation is limited to the captured FabuBlox simulation interface.

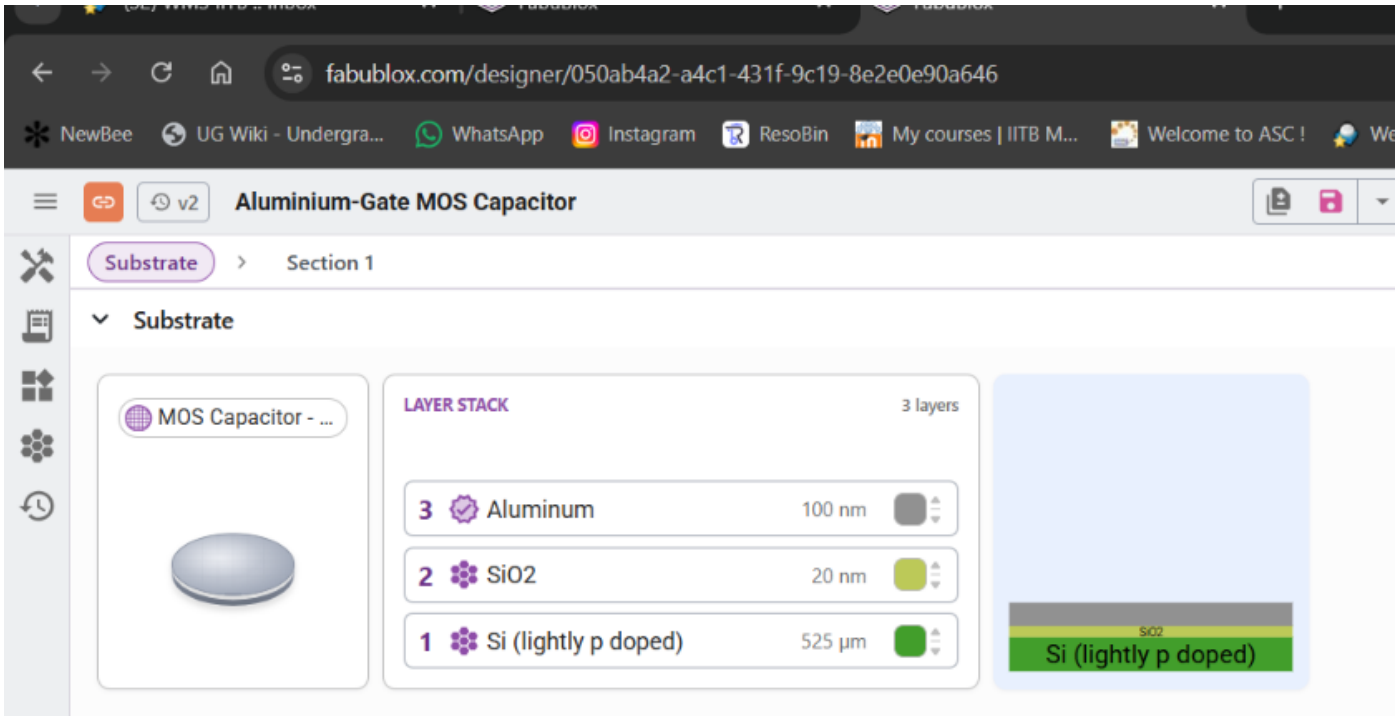


Figure 2. Saved project title and layer stack: lightly p-doped Si, 20 nm SiO2, and 100 nm aluminium.

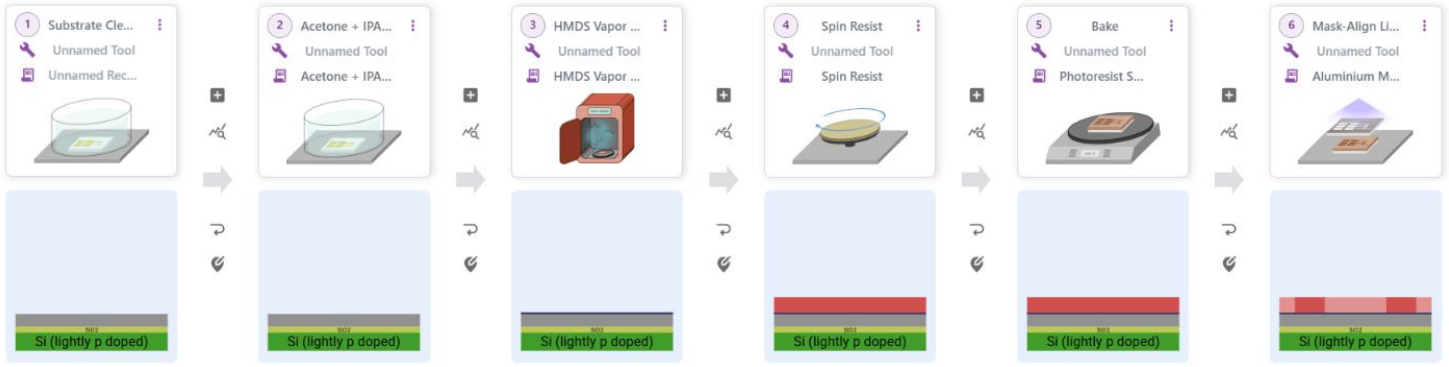


Figure 3. Workflow steps 1-6: substrate clean through mask-align lithography.

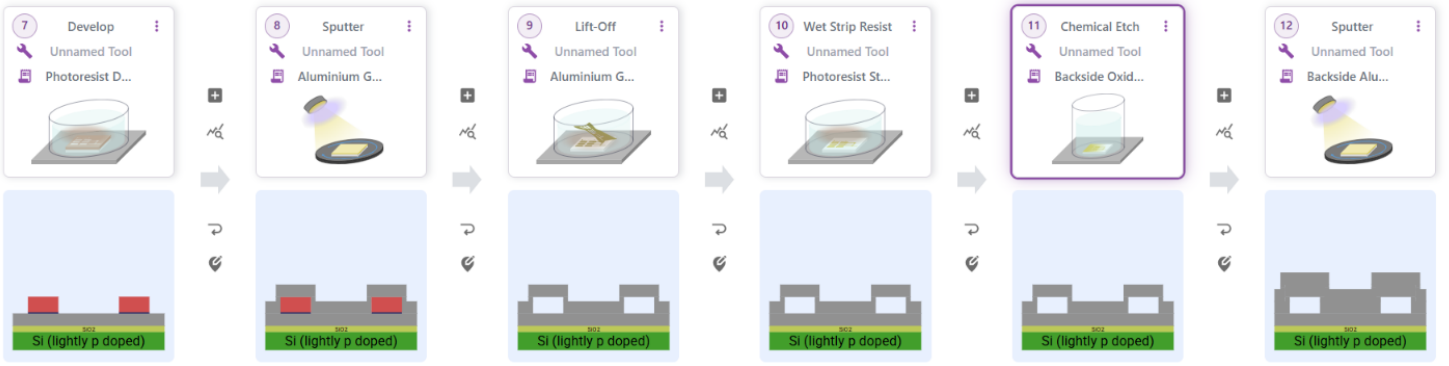


Figure 4. Workflow steps 7-12: develop through backside aluminium sputter.

4. Simulation evidence and limitations

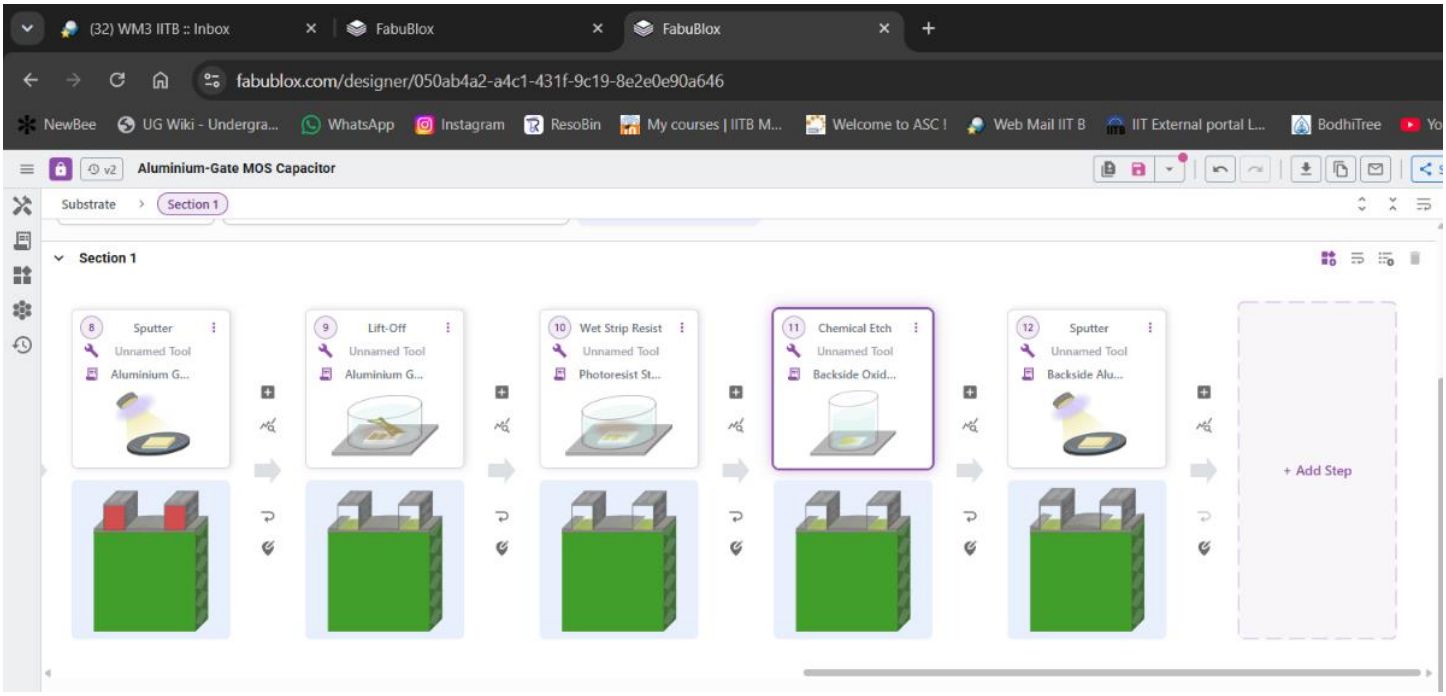


Figure 5. Final FabuBlox simulation view. The notices pane displays 'No notices.'

The screenshot confirms the interface did not display notices in that final captured view. It does not confirm a process recipe is physically validated, that a device was fabricated, or that device performance has been measured.

The editor did not provide suitable annealing or electrical-characterization steps for this flow. Those activities are therefore not represented as completed. If the project progresses, the appropriate next work is team review of approved process conditions, a safe fabrication plan, optical inspection, and only then electrical measurements using approved equipment.

5. References

See REFERENCES.md for the full reference list. Key background sources are Sze and Ng, *Physics of Semiconductor Devices*, and Wolf and Tauber, *Silicon Processing for the VLSI Era*.